GF(27) in GALG and FFGFT:

Why G(3) Contains No GF(27) but G(6) Does

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GF(27) in GALG and FFGFT

Abstract

Doug Matzke reports in his IPI mail of 1 September 2026 that a systematic search in GALG G(3) (6561 attempts) finds no element of order 26. This document explains algebraically why, shows where such elements exist in G(6), and draws the consequences for the FFGFT mass layer.

Three theorems: **(A)** G$(3) \cong M_2($GF$(9))^2$: order 26 is algebraically impossible in G(3) — not a search error, but a theorem. **(B)** G$(6) \cong M_8($GF$(9))$: order-26 elements exist; a 5-blade example can be verified directly in GALG. **(C)** GF(27) is present in G(6) as an order structure, not as a subfield (since $3 \nmid 8$) — exactly the FFGFT statement from Doc. 338.

Consequence: $1/\alpha = 3700/27$ lives on the G(3)/GF(9) level; the lepton mass layer $(m_\mu/m_e)^2 = 43200$ needs G(6) with order-26 elements. Verification script: `pruef_341_gf27_in_galg.py` (all assertions **[B]**).

.1 Background and Notation

This section provides the necessary background for readers not immediately familiar with all the algebraic structures involved. It can be skipped by specialists.

Status markers

This document uses the standard FFGFT corpus status markers:

- **[K]** — *Confirmed*: algebraically or numerically fully proved; all assertions in the verification script passed.
- **[B]** — *Established*: numerically secured by sampling or explicit construction; analytic proof present or straightforward.
- **[S]** — *Speculative*: indication or conjecture, not yet fully proved.

Galois fields GF(p^n)

A *Galois field* GF(p^n) is the unique finite field with p^n elements, where p is prime (the *characteristic*) and $n \geq 1$. Its multiplicative group GF$(p^n)^* =$ GF$(p^n) \setminus \{0\}$ is cyclic of order $p^n - 1$.

- GF(3): the integers $\{0, 1, 2\}$ modulo 3. Multiplicative group order 2.
- GF$(9) =$ GF(3^2): elements $a + bi$ with $a, b \in \{0, 1, 2\}$ and $i^2 = -1 \equiv 2$. Multiplicative group order 8. This is Doug's symmetric Z3C.
- GF$(27) =$ GF(3^3): 27 elements, multiplicative group order $26 = 2 \cdot 13$.
- GF$(81) =$ GF(3^4): 81 elements, multiplicative group order $80 = 2^4 \cdot 5$.

The *order* of an element x in a group is the smallest positive k with $x^k = 1$. If the group has order m, then every element order divides m (Lagrange's theorem). This is the key fact used throughout.

Clifford algebra Cl(n) and GALG G(n)

The Clifford algebra Cl(n) is generated by n basis vectors $e_1, \ldots, e_n$ with the rule $e_i^2 = +1$ (hyperbits in Matzke's language) and $e_i e_j = -e_j e_i$ for $i \neq j$. A *blade* of grade r is a product $e_{i_1} e_{i_2} \cdots e_{i_r}$ with $i_1 < i_2 < \cdots < i_r$; together with the scalar 1 (grade 0) they form a basis of 2^n elements. *GALG G*(n) is Cl(n) over the coefficient field GF$(9) =$ Z3C (i.e. each blade carries a Z3C = GF(9) coefficient). As a matrix algebra: Cl$(3) \cong M_2($GF$(9))^2$ and Cl$(6) \cong M_8($GF$(9))$, where $M_k(F)$ denotes $k \times k$ matrices over the field F.

Why GF(27) and order 26?

The lepton mass ratio $(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 5^2 \cdot 27$ involves $27 = |$GF$(27)|$ (Doc. 338, pruef_330/331). The multiplicative group GF$(27)^*$ has order $26 = 2 \cdot 13$. An element of order 26 in any algebra generates a copy of GF$(27)^*$. Doug's search for order-26 elements in G(3) is therefore a direct test of whether GF(27) structure is present in G(3).

Companion matrix

Given a monic polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, its *companion matrix* is the $n \times n$ matrix with 1's on the subdiagonal and the coefficients $-a_0, \ldots, -a_{n-1}$ in the last column. Its characteristic polynomial is $f(x)$, so its eigenvalues are the roots of f and its order in GL_n equals the multiplicative order of the roots.

.2 Occasion and Starting Point

Doug Matzke (IPI, 1 Sept. 2026) reports finding in GALG G(3): GF(9) in the generator $(a+ja)^8 = 1$; GF(81) in G(3) with $(1+a+b+c+(a\wedge b))^{80} = 1$; and element orders $\{2, 3, 4, 8, 16, 32, 40, 80\}$ — all of which divide $80 = |\mathrm{GF}(81)^*|$, consistent with the $\mathrm{Cl}(3) \cong M_2(\mathrm{GF}(9))^2$ structure where eigenvalues live in GF(81).

An explicit search for elements of order 26 (which would signal GF$(27)^*$ structure):

```
gasolve([a,b,c], is_root26): Attempted 6561 with 0 found.
```

$6561 = 3^8$ is the number of elements in a 4-trit search space — exhaustive at that depth.

This is not a search error or a limitation of the tool. It is an algebraic theorem: order 26 requires the prime factor 13, and 13 does not appear in $80 = |\mathrm{GF}(81)^*|$. The following sections prove this precisely and show where GF(27) structure does appear — in G(6), not G(3).

.3 Theorem A: G(3) Contains No GF(27)

Structure of G(3)

GALG G(n) = Cl(n) over the symmetric Z3C = GF(9) = GF$(3)[i]$, where $i^2 = -1$, characteristic 3. For odd $n = 3$:

$$G(3) = \mathrm{Cl}(3) \cong M_2(\mathrm{GF}(9)) \oplus M_2(\mathrm{GF}(9)) \tag{1}$$

Every irreducible representation is 2-dimensional over GF(9).

Eigenvalue Structure

The eigenvalues of a 2×2 matrix over GF(9) lie in the algebraic closure — in the smallest extension field GF(9^2) = GF(81).

$$|\mathrm{GF}(81)^*| = 80 = 2^4 \cdot 5 \tag{2}$$

The number 13 does not divide 80. Therefore:

- No element of $M_2(\mathrm{GF}(9))$ has an order divisible by 13.
- Order 26 requires $13 \mid \mathrm{ord}(X)$ — impossible in G(3).
- Doug's search space $\{a, b, c\}$ spans G(3) — the result 0/6561 is algebraically forced **[B]**.

Consistency with Doug's Findings

Table 1: Orders in GALG G(3) and their Galois origin

Order	Origin	Divides
2	$\lvert$GF$(3)^*\rvert$	80
4	$\varphi(5) = 4$	80
5	min. prime factor $\lvert$GF$(81)^*\rvert$	80
8	$\lvert$GF$(9)^*\rvert$	80
16, 20, 40, 80	subgroups of GF$(81)^*$	80
3	unipotent (char. 3)	—
26	$\lvert$GF$(27)^*\rvert$, needs factor 13	**not** 80

A numerical sample of 3000 random Cl(3) elements confirms: no order is divisible by 13 **[B]**.

.4 Theorem B: G(6) Contains Order-26 Elements

Structure of G(6)

$$G(6) = \mathrm{Cl}(6) \cong M_8(\mathrm{GF}(9)) \tag{3}$$

Irreducible representations are 8-dimensional over GF(9).

Construction via Irreducible Polynomial

The polynomial $f(x) = x^3 + 2x + 1$ over GF(3) (equivalent to $x^3 - x + 1$) has no root in GF(3) and is therefore irreducible. Since $\gcd(3, 2) = 1$, f remains irreducible over GF(9); its roots lie in GF$(27) \setminus$ GF(3). The companion matrix

$$C_f = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\mathrm{GF}(3)) \tag{4}$$

has order 26 in GL$_3$(GF(3)) (f is primitive) **[B]**.

Embedding in G(6) as a GALG Element

The 8×8 element $X = C_f \oplus C_f \oplus (-I_2)$ satisfies:

- $X^{26} = 1$, $X^{13} \neq 1$, $X^2 \neq 1$ — order exactly 26 **[B]**
- Blade decomposition over 38 Z3C blades from $\{e_1, \dots, e_6\}$

Minimal Example: 5 Blades, Directly Verifiable in GALG

A systematic search shows: order-26 elements need at least 5 blades; with 4 blades there are none. The following 5-blade element can be verified directly in Doug's GALG tool:

$$X = -(1+i)(e_1e_2) + (1+i)(e_1e_2e_3e_6) + (1-i)(e_1e_2e_5e_6) + (1+i)(e_1e_4e_5e_6) - (e_1e_2e_3e_4e_6) \tag{5}$$

GALG syntax:

```
X = -(1+1j)*(e1^e2) + (1+1j)*(e1^e2^e3^e6)
+ (1-1j)*(e1^e2^e5^e6) + (1+1j)*(e1^e4^e5^e6)
- (e1^e2^e3^e4^e6)
X**26 == 1 # True
X**13 == 1 # False
```

[B] (pruef_341, all assertions passed)

Frequency in G(6)

Table 2: Fraction of elements with $13 \mid$ ord(X) in G(6) by blade support

Support (number of blades)	Hits / Invertible	Fraction
≤ 4	0 / ~450	0 %
5	rare	< 1 %
6	85 / 478	17.8 %
10	190 / 511	37.2 %

Order-26 elements are common in G(6) — Doug's search in G(3) (span$\{a, b, c\}$) could not find them for two reasons: algebraically (Theorem A) and because G$(3) \subsetneq G(6)$.

.5 Theorem C: GF(27) as Order Structure, not Subfield

Subfield Criterion

A unital embedding GF$(27) \hookrightarrow M_n(GF(9))$ makes GF$(9)^n$ a GF$(27) \otimes$ GF(9) = GF(729)-vector space. Since [GF(729) : GF$(9)] = 3$, we need $3 \mid n$.

Table 3: GF(27) as subfield in $M_n($GF$(9))$

n	$3 \mid n$	Subfield possible?
2	no	no (G(3))
8	no	no (G(6))
3	yes	yes
6	yes	yes
9	yes	yes

For G(6) with $n = 8$: no GF(27) subfield. But the minimal polynomial of X factors:

$$\chi_X(x) = (x^3 + 2x + 1)^2 \cdot (x + 1)^2 \quad \in \mathrm{GF}(3)[x] \tag{6}$$

Two cubic factors — eigenvalues in GF(27) — and one linear factor from GF(3). GF(27) appears as splitting field, not as subalgebra.

Connection to Docs. 330/338

This is exactly the FFGFT statement from Docs. 330 and 338:

$$\mathrm{GF}(9) \not\subset \mathrm{GF}(27) \quad (\text{since } \gcd(2, 3) = 1) \tag{7}$$

Both fields sit in parallel in GF(3^6) = GF(729), not in a tower. The mass-ratio identity $(m_\mu/m_e)^2$ = $|$GF$(9)^*|^2 \cdot 5^2 \cdot |GF(27)|$ uses both fields *in parallel* — consistent with Theorem C.

.6 Vacuum Structure and Z3 Separation (pruef_341_vakuum_witt_z3)

Doug's IPI mail of 1 September 2026 (22:05) not only verifies the order-26 element but discloses the full structure of six vacuum states Vss$[0\ldots5]$ in G(6) and their grade-2 (bivector) content. The following results are verified in `pruef_341_vakuum_witt_z3.py`.

Witt Pairs and the Number Operator N

The three grade-2 bivectors

$$B_1 = -i(e_1e_3), \quad B_2 = -i(e_2e_6), \quad B_3 = -i(e_4e_5) \tag{8}$$

satisfy $B_k^2 = +1$, commute pairwise, and commute with jE_7. They form the three *Witt pairs* $\{13\}, \{26\}, \{45\}$ — the decomposition of Cl(6) into three complex planes (Fock space of 3 modes). The number operator $N = B_1 + B_2 + B_3$ is simultaneously the grade-2 part of Vss$[0]$ and its constructor **[B]**:

$$(1+N)(1 \pm jE_7) = \text{Vss}[0] \quad \text{(sign depends on basis orientation of } E_7) \tag{9}$$

Triality Built into Z3C

Over $\mathbb{C}$, the Fock number operator $N = B_1 + B_2 + B_3$ would have eigenvalues $\{-3, -1, +1, +3\}$. Over GF(9) in characteristic 3: $-3 \equiv 0 \equiv +3$ **[B]**.

$$N \bmod 3 \in \{0, +1, -1\} \quad \text{with multiplicities} \quad \{2, 3, 3\} \tag{10}$$

The vacuum ($n = 0$) and the fully occupied state ($n = 3$) *coincide* in characteristic 3. The $\mathbb{Z}_3$ cycle is closed without any extra input. This is exactly the FFGFT sector variable $\Delta w = N \bmod 3$ (Doc. 336, $Q_{\mathbb{Z}_3} = N \bmod 3$). The $+1$-eigenspace is 3-dimensional — Doug's fermion-generation candidate.

Z3 Separation of Even and Odd Vacuums

The cyclic Witt-pair rotation

$$\sigma: \quad e_1 \to e_2 \to e_4 \to e_1, \quad e_3 \to e_6 \to e_5 \to e_3 \tag{11}$$

cycles $B_1 \to B_2 \to B_3 \to B_1$ while leaving E_7 invariant (even permutation). Applied to Doug's six vacuums **[B]**:

$$\begin{aligned} &\sigma(\text{Vss}[0]) = \text{Vss}[0], \quad \sigma(\text{Vss}[2]) = \text{Vss}[2], \quad \sigma(\text{Vss}[4]) = \text{Vss}[4] \\ &\sigma(\text{Vss}[1]) = \text{Vss}[3], \quad \sigma(\text{Vss}[3]) = \text{Vss}[5], \quad \sigma(\text{Vss}[5]) = \text{Vss}[1] \end{aligned} \tag{12}$$

Even vacuums $[0, 2, 4]$: individual $\mathbb{Z}_3$-fixed points.
Odd vacuums $[1, 3, 5]$: a single $\mathbb{Z}_3$-orbit of three.

This algebraic separation is the *same* as the Frobenius fixed-point/orbit split from Doc. 339: fixed points = massive sector (fermion generations), orbit = other sector (Doug's dark energy/mass candidate). Doug finds this split via grade parity; FFGFT finds it via Frobenius eigenstructure. **They agree [B].**

Bilaterals as Sector-Changers (Gluon Role)

The Witt raisers $ap_k = (e_a + i\, e_b)/2$ for each pair satisfy **[B]**:

$$[N,\ ap_i \cdot ap_j] = +1 \cdot (ap_i \cdot ap_j) \tag{13}$$

The bilaterals $ap_i \cdot ap_j$ shift the N-sector by ± 1 mod 3, i.e. they are Δw-changers. This is the gluon role in Doc. 339 (three-fold orbit transmutation between sectors). Doug's question "how do gluons transmute quarks using bilaterals?" is answered: they shift the winding-mode sector Δw by ± 1.

Projector Properties

$$\text{Vss}[0]^2 = -\text{Vss}[0], \qquad N \cdot \text{Vss}[0] = 0 \tag{14}$$

$-\text{Vss}[0] = 2 \cdot \text{Vss}[0]$ is idempotent over GF(9); Vss$[0]$ projects onto the $N = 0$ sector (vacuum + fully occupied state).

.7 Higher Roots of Unity: root7, p27 Projectors, Coefficient Purity

Doug's follow-up questions (IPI, 4–7 Sept. 2026) concern 7th roots of unity, projectors with $X^{27} = X$, and a structural objection to the order-26 element. Verified in `pruef_341_primordnung.py`, `pruef_341_root7_gf9.py`, `pruef_341_r26_rein_p27.py`.

Entry Order of Primes

The smallest k with $p \mid 3^k - 1$ (i.e. $k = \text{ord}_p(3)$) gives the first field GF(3^k) in which p-th roots of unity exist **[K]**:

p	k	$\lvert\text{GF}(3^k)^*\rvert$
13	3	$26 = 2 \cdot 13$
5	4	$80 = 2^4 \cdot 5$
11	5	$242 = 2 \cdot 11^2$
7	6	$728 = 2^3 \cdot 7 \cdot 13$

Within G(3) (eigenvalues in GF(81)) root5 is indeed the first odd-prime root; globally 13 enters before 5. In no GF$(3^k)^*$ is there an element of order 3, since $3^k - 1 \equiv 2 \pmod 3$ for all k: tripotents $t^3 = t$ with $t \neq 0, \pm 1$ are a matrix phenomenon, never a field phenomenon **[K]**.

root7 Exists Only in GF(9)-Arithmetic

$7 \nmid |\text{GF}(9)^*| = 8$, and $7 \mid 3^k - 1$ first at $k = 6$. A genuine complex 7th root of unity has coefficients in $\mathbb{Q}(e^{2\pi i/7})$ (degree 6 over $\mathbb{Q}$); cos$(2\pi/7)$ has minimal polynomial $8x^3 + 4x^2 - 4x - 1$ and lies neither in $\mathbb{Q}(i)$ nor in GF(9). Constructions of the form $\cos\theta + B\sin\theta$ (Euler form over $\mathbb{R}$) are therefore not GALG elements **[K]**.

What does exist is an order-7 element in G(6) over GF(9): $\Phi_7(x) = x^6 + \cdots + 1$ is irreducible over GF(3); its companion matrix C has order 7 in GL$_6$(GF(3)), and $X_7 = C \oplus I_2 \in M_8(\text{GF}(9))$ has order 7 with 47 blades over all grades, all coefficients in $\{\pm 1, \pm i\}$ **[K]**. A compact alternative: take any Y with $7 \mid \text{ord}(Y)$ and set $X = Y^{\text{ord}(Y)/7}$. Example **[B]**:

```
Y = (1j)*(e1^e2^e3) - (e1^e3^e6) + (e2^e3^e4) - (e2^e4^e6)
- (1j)*(e4^e5^e6) - (e1^e2^e4^e5)
ord(Y) = 182 = 2*7*13;  X = Y**26;  X**7 == 1, X != 1
```

X has 10 blades, all coefficients pure (± 1 or $\pm i$).

p27 Projectors as a Field-Tower Hierarchy

Since $x^{27} - x = \prod_{a \in \mathrm{GF}(27)}(x-a)$, the condition $X^{27} = X$ holds exactly when X is semisimple with spectrum in GF(27). This generalises tripotents ($X^3 = X$, spectrum in GF(3)) along the field tower **[K]**:

	$X^3=X$	$X^9=X$	$X^{27}=X$	$X^{81}=X$	$X^{729}=X$
N (tripotent)	✓	✓	✓	✓	✓
r_{26} (order 26)	–	–	✓	–	✓
X_7 (order 7)	–	–	–	–	✓

Doug's "power-of-27 projector" is thus precisely: spectrum in GF(27). X_7 needs GF(729) because $7 \nmid 26$.

Coefficient Purity and the Order-26 Element

Doug objects that r_{26} contains $-(1+j)(e_1e_2)$, mixing e_1e_2 (square -1) with je_1e_2 (square $+1$), which he reads as two signature choices. Algebraically, $1+j$ is simply the generator of GF$(9)^*$ (order 8, norm -1) — Doug's own $(a+ja)^8 = 1$ — so $-(1+j)B$ is scalar times blade. The objection is a *physical* purity requirement: each blade coefficient in $\{\pm 1\}$ or $\{\pm j\}$, never $a+bj$ with $a, b \neq 0$. That requirement is satisfiable **[B]**: about 2.5 % of random 5-blade elements with pure coefficients have order exactly 26, e.g.

```
X = (1j)*e3 + (1j)*(e2^e5) + (1j)*(e3^e6) + (e4^e6) + (e3^e5^e6)
X**26 == 1,  X**13 != 1
```

X_7 above is already pure (no mixed coefficient).

A Convention Note: the Sign of E_7

GALG defines $E_7 = -(e_1 \wedge \cdots \wedge e_6)$. An element transmitted with a term "$(+i)\,E_7$" meaning $+i\,e_1 \cdots e_6$ is read in GALG as $-i\,e_1 \cdots e_6$. For X_7 this sign flip yields a non-invertible element with $X^{27} = X$ and no $X^n = 1$ — exactly the behaviour Doug observed **[K]**. With Doug's E_7 the last term of X_7 must be entered as `(-1j)*E7`.

.8 Consequences for FFGFT and GALG

Table 4: GF(27) structure and vacuum separation in GALG and FFGFT

Level	GALG	FFGFT	Status
α	G(3)/GF(9)	$1/\alpha = 3700/27$, cancels	**[B]**
Lepton masses	G(6), ord. 26 needed	$(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 5^2 \cdot 27$	**[K]**
GF(27) subfield	absent in G(3), G(6)	parallel in GF(729)	**[B]**
Vacuum split	Vss$[0,2,4]$ vs. Vss$[1,3,5]$	massive vs. other sector	**[B]**
$\mathbb{Z}_3$ structure	fixed pts vs. orbit	Frobenius split (Doc. 339)	**[B]**
N eigenvalues	$\{0,+1,-1\}$, mult. $\{2,3,3\}$	$\Delta w \in \{0,+1,-1\}$	**[B]**
Bilaterals	$[N, ap_iap_j] = +1$	Δw-changer, gluon role	**[B]**

The fine-structure constant $1/\alpha = 3700/27$ lives on the GF(9)/G(3) level: $|\mathrm{GF}(27)| = 27$ cancels in the derivation. The lepton mass layer and the vacuum structure both need G(6). Doug's two-family split of vacuums and FFGFT's Frobenius separation find the same $\mathbb{Z}_3$ orbit structure from different starting points **[B]**.

.9 Theorem D: Vacuum Structure and $\mathbb{Z}_3$ Orbit Decomposition

The six vacuum states Vss$[0, \ldots, 5]$ split under the $\mathbb{Z}_3$ rotation σ (Doc. 339) into two classes **[B]**:

- **Even vacua** Vss$[0, 2, 4]$: σ-fixed points (massive sector); $N \bmod 3 = 0$ (multiplicity 2).
- **Odd vacua** Vss$[1, 3, 5]$: $\mathbb{Z}_3$ three-cycle (other sector); $N \bmod 3 \in \{+1, -1\}$ (multiplicity 3 each).

The bilateral products $ap_i \cdot ap_j$ shift the N-sector by ± 1 mod 3 — they are Δw-changers and play the gluon role of Doc. 339 **[B]**. Doc. 344 (Theorem F) shows: this decomposition agrees exactly with the tripotent property of the even vacua and the GALG grade construction — two independent derivations of the same structure.

Appendix: For the Reader Without an Algebra Background

This appendix explains the key ideas in plain language, assuming no prior knowledge.

What is a finite field?

A field is a set of numbers in which you can add, subtract, multiply and divide (except by zero). Familiar examples are the real numbers $\mathbb{R}$ and the rational numbers $\mathbb{Q}$. A *finite* field has only finitely many elements. You compute "modulo" a prime p: discard all multiples of p. Example GF(3): the numbers $\{0, 1, 2\}$ with $1 + 2 = 0$, $2 + 2 = 1$, etc.

What does GF(p^n) and GF$(p^n)^*$ mean?

GF(p^n) is the unique finite field with exactly p^n elements. The *star* GF$(p^n)^*$ denotes all elements *except zero* — analogous to $\mathbb{Z}^* = \{\ldots, -1, +1, \ldots\}$ vs. $\mathbb{Z}$:

$$|\mathrm{GF}(27)| = 27 \qquad |\mathrm{GF}(27)^*| = 26 = 27 - 1$$

Group order and Lagrange's theorem

GF$(p^n)^*$ is always a cyclic group of order $p^n - 1$. The *order* of an element x is the smallest k with $x^k = 1$. By Lagrange's theorem, every element order *divides* $p^n - 1$. Think of a clock hand that returns every 4 hours: it cannot fit on a 6-hour clock because 4 does not divide 6.

Concretely:

- GF$(81)^*$: order $80 = 2^4 \cdot 5$. Prime factors: 2 and 5 only. Order 26 $= 2 \cdot$ **13** is impossible — 13 does not divide 80.
- GF$(27)^*$: order $26 = 2 \cdot 13$. Elements of order 13 and 26 exist here.

Why does GF(27) appear in the lepton masses?

$(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 25 \cdot 27$. The 27 is |GF$(27)$| (total field size including zero); the 8 is |GF$(9)^*$| (group order of GF(9)). These arise because the FFGFT winding numbers on the T^4 torus are algebraically linked to these fields.

Why does Doug find no GF(27) in G(3)?

G(3) uses 2×2 matrices over GF(9); eigenvalues live in GF(81). GF$(81)^*$ has order 80, and $13 \nmid 80$, so order 26 is impossible. In G(6) with 8×8 matrices, order 26 is achievable. A 5-blade example is given in §.4 and directly verifiable in GALG.

Verification Script

`pruef_341_gf27_in_galg.py` — Theorems A, B, C verified numerically (Cl(6) over GF(9) as 8×8 matrices; exact order determination; blade decomposition with Z3C coefficients; samples of 3000 and 600 elements). All assertions passed **[B]**.

`pruef_341_primordnung.py` — entry order of primes ord$_p(3)$; $3^k - 1 \equiv 2 \pmod 3$ (no field tripotents); trine-product counterexample in M_2(GF(3)).

`pruef_341_root7_gf9.py` — non-existence of root7 with Z3C coefficients; order-7 element $X_7 = C_{\Phi_7} \oplus I_2$ over GF(9); contrast with $\cos\theta + B\sin\theta$.

`pruef_341_r26_rein_p27.py` — $1 + j$ as GF$(9)^*$ generator; pure-coefficient root26; $X^q = X \Leftrightarrow$ spectrum in GF(q); E_7 sign discrepancy; compact root7 Y^{26}.

`pruef_341_vakuum_witt_z3.py` — Vacuum structure verified: constructor identity, Witt pairs, N eigenvalues, $\mathbb{Z}_3$ separation of Vss$[0\ldots5]$, bilaterals as sector-changers, projector properties. All assertions passed **[B]**.

Note on Computational Tools

This document is the work of Johann Pascher and Doug Matzke. Johann Pascher brings years of theoretical development in FFGFT (documented in over 350 corpus documents), poses the research questions, reads and corrects all results, and draws the physical conclusions. Doug Matzke brings the GALG framework, its implementation, and the real-time verification of every algebraic claim in his own tool. The correspondence between both authors drives the direction of inquiry.

Claude (Anthropic) was used as a tool for several tasks that are impractical to do by hand:

- **Computation:** Writing and executing the Python verification scripts (`pruef_341_*`); performing randomised blade searches (thousands of candidates per run); computing order determinations that require iterated 8×8 matrix multiplication mod 3 over exponent spaces up to $\sim 10^{30}$; computing blade decompositions.
- **Writing and typesetting:** Drafting the LaTeX source of this document from the authors' results, notes, and corrections; maintaining the corpus file structure, changelogs, and register entries; preparing correspondence drafts that Johann Pascher reviews and approves.
- **Iteration:** Incorporating corrections, checking consistency across the corpus, and updating cross-references — tasks that scale with the size of a 350-document corpus.

The tool executes; the authors decide what to compute and write, judge every result, correct errors, and determine what the results mean physically.

A central methodological concern is the prevention of fabricated content. The workflow enforces this at two levels: (1) every algebraic claim is backed by a Python verification script with explicit assertions — if an assertion fails, the claim is not used; (2) targeted instructions direct the tool to read the existing corpus before writing, never to introduce facts not present in the sources, and to mark all results explicitly as confirmed (**[K]**), established (**[B]**), or speculative (**[S]**). Invented content is the primary risk when using language models for scientific work; the scripts and instructions are designed to make it detectable and avoidable.

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